
Statistics 8

Spring - 2026

Class Meeting Day(s), Time, and Location: Information Located on UCI Class Schedule

<https://www.reg.uci.edu/perl/WebSoc>

Learning Mode: (classroom)

Instructor: Jessalyn Sebastian

Email: jessalys@uci.edu

Office Location: DBH 2048

Office hours:

MW: 2:00 – 3:00 pm

I. Course Description*

Introductory statistical techniques used to collect and analyze experimental and observational data from health sciences and biology. Includes exploration of data, probability and sampling distributions, basic statistical inference for means and proportions, linear regression, and analysis of variance.

II. Student Learning Outcomes*

By the end of this course, you should be able to analyze and present data, design observational and experimental studies, use probabilities to model and predict random events, and use inference procedures to test hypotheses and estimate population parameters to reach conclusions in context. What you learn in this class should help you understand broadly the methodology, results, and issues of studies presented in your other classes or in news stories. I also hope that you will come to appreciate statistics as a cool and really interesting subject!

III. How to Succeed in the Course

To succeed in this course, please come to class and discussion prepared and on time, ask questions when you need help, attend office hours, and actively participate in in-class discussions.

IV. Required Materials*

Mind On Statistics, 6th edition with Webassign by Jessica Utts and Robert Heckard.

You need the Webassign homework access.

Scientific Calculator (not graphing calculator) for exams.

V. Course Technology Requirements*

We are using free statistical software called R. You can download these onto your own computer (PC and Mac) or use web-based REPLs (linked in Canvas). Instruction will be provided in class and/or in discussion sections. The programs are available in the ICS computer labs as well, but to use them there you need to set up an account with ICS. Instructions for doing so are provided at this link: http://www.ics.uci.edu/~lab/students/acct_activate.php.

VI. Communication Expectations*

Please contact the instructor and TAs via email.

VII. Assignment Details*

- **Discussion Assignments** – These assignments will be completed during discussion section. They will be short worksheets based on the material from the prior week, to be completed in groups.
- **Lecture attendance** – Lecture attendance will be recorded via PollEverywhere, so you will be expected to bring a device that can connect to the internet. Lecture attendance will not be a part of the grade breakdown; instead, if you attend at least 23 lectures (excluding the 2 midterms and the first lecture), your grade will be automatically rounded up if it is within 0.5% of the next letter grade higher. No other rounding requests will be considered.
- **Homework** – These assignments are an opportunity to see how the various concepts connect and to develop a deeper understanding of the overall material.
- **Midterms (2)** – There will be two midterms, taken during lecture in person and they are intended to help students practice and master core concepts. They serve as checkpoints for your progress and preparation for the cumulative final exam.
- **Final Exam** – The final is comprehensive and may include any topic covered throughout the course. You should review all prior materials and aim to synthesize your learning into a strong overall understanding.

Late and Incomplete Assignments

There are no extensions for late work.

Academic Dishonesty

Students are intellectually and morally obliged to pursue their course of study with honesty and integrity. Therefore, in preparing and submitting course materials and exams, a student shall not yield to cheating or plagiarism.

Plagiarism is defined as the act of incorporating ideas, words, or specific substance of another and submitting as one's own work to fulfill academic requirements without giving credit to the appropriate source. Plagiarism includes but is not limited to:

- Submitting work, either in part or in whole, completed by another
- Omitting footnotes for ideas, statements, facts, or conclusions that belong to another
- Omitting quotation marks when quoting directly from another, whether it be a paragraph, sentence, or part thereof
- Paraphrasing in a close or lengthy manner of the writings of another
- Submitting another person's artistic works, such as musical compositions, photographs, paintings, drawings, or sculptures
- Submitting as one's own work papers purchased from research companies.

Using AI-generated content as a replacement for original thoughts and viewpoints is discouraged, and students are urged to approach their projects and write with creativity and originality. While using AI tools (i.e. ChatGPT, Bard) may be appealing, it is crucial to form and communicate one's ideas because this is what academic writing is all about.

VIII. Course Schedule/Calendar*

Weekly Class Content & Due Dates for Assignments		
Week #	Class (Topics) Approximate	Assignment Details & Due Dates
One (1)	Topic: Welcome & study designs, bias, experiments and observational studies	Friday – HW0 due (Google form submission)
Two (2)	Topic: Descriptive statistics (categorical and quantitative variables)	Wednesday – Discussion work due Friday – HW1 due
Three (3)	Topic: Association (categorical and quantitative variables)	Wednesday – Discussion work due Friday – HW2 due
Four (4)	Topic: Probability concepts	Monday – Midterm 1 in class Wednesday – Discussion work due Friday – HW3 due
Five (5)	Topic: Random variables and probability distributions	Wednesday – Discussion work due Friday – HW4 due
Six (6)	Topic: Sampling distributions, CLT, statistical inference concepts	Wednesday – Discussion work due Friday – HW5 due
Seven (7)	Topic: Statistical inference concepts, confidence interval for a mean	Wednesday – Midterm 2 in class Friday – HW6 due
Eight (8)	Topic: Statistical inference for one population	Wednesday – Discussion work due Friday – HW7 due
Nine (9)	Topic: Statistical inference for multiple populations	Wednesday – Discussion work due Friday – HW8 due
Ten (10)	Topic: Exam review	Wednesday – Discussion work due Friday – HW9 due
Finals Week	In Class Final Exam Date and time found on UCI Class Schedule	In Class Final Exam

IX. Grading Methods*

Overview and Breakdown		
Assignment	Submission Method	Percentage
<i>Discussion Assignments</i>	<i>Canvas</i>	<i>10%</i>
<i>Lecture Attendance</i>	<i>PollEverywhere</i> If you attend at least 23 lectures (excluding the midterm lectures and the first lecture), your grade will be automatically rounded up <i>if it is within 0.5% of the next letter grade higher.</i>	<i><0.5%</i>
<i>Homework</i>	<i>WebAssign</i>	<i>10%</i>
<i>Midterms</i>	<i>2 Midterms (April 20, May 13)</i> Need: valid ID, scientific calculator	<i>20% each</i>
<i>Final Exam</i>	<i>In class, during finals week</i> Need: valid ID, scientific calculator Your final exam score may replace the lower of the two midterm scores if it is higher.	<i>40%</i>
	<i>Total</i>	<i>100%</i>

Grade Scale

A+	[97.5% - 100%]	C+	[77.5% - 80%]
A	[93.5% - 97.5%)	C	[73.5% - 77.5%)
A-	[90.0% - 93.5%)	C-	[70.0% - 73.5%)
B+	[87.5% - 90%]	D+	[67.5% - 70%]
B	[83.5% - 87.5%)	D	[63.5% - 67.5%)
B-	[80.0% - 83.5%)	D-	[60.0% - 63.5%)
		F	[0% -60.0%)

X. Student Resources

Example Campus Resources

- [UC Policy on Sexual Violence and Sexual Harassment](#)
- [UCI Library](#)
- [UCI Center for Excellence in Writing and Communication](#)
- [The Learning & Academic Resource Center \(LARC\)](#)
- [UC Irvine Launches Online Learning Research Center](#)
- [UCI Student Portal](#)
- [UCI Disability Service Center \(DSC\)](#)
- [Office Information Technology \(OIT\)](#)
- [UCI Wellness, Health, and Counseling](#)
- [UCI Office of Academic Integrity & Student Conduct](#)
- [UCI Counseling Center](#)
- [Student Affairs](#)

Basic Needs Security

Any student who faces challenges securing sufficient food or housing and believes this may affect their performance in the course is urged to contact the Dean of Students for support ([A Message from the Dean of Students](#); 949-824-5181). Also, note the following resources available at UCI:

- [Student Outreach and Retention Center \(SOAR\)](#)
- [UCI Basic Needs Center](#) – located at the FRESH Basic Needs Hub at 4079 Mesa Rd (Lot 5 trailers), open M-Th 11am - 5pm

UC Policy on Sexual Violence and Sexual Harassment

Title IX prohibits gender discrimination, including sexual harassment, domestic and dating violence, sexual assault, and stalking. If you have experienced sexual harassment or sexual violence, you can receive confidential support and advocacy at the [Campus Advocacy Resources & Education \(CARE\)](#) office, by phone at (949) 824-7273, or by email at care@uci.edu. We encourage you to also report incidents through the [Office of Equal Opportunity and Diversity's \(OEOD's\) reporting page](#).

In addition, the campus Title IX Officer, along with the OEOD, has provided a number of local and national [Sexual Violence Resources](#) should you need immediate assistance related to sexual violence or other sexual misconduct. These resources are available even when the campus is closed or on an administrative holiday.

Faculty and TAs are required under the [UC Policy on Sexual Violence and Sexual Harassment](#) to inform the Title IX Office should they become aware that you or any other student has experienced sexual violence or sexual harassment.

XI. Course Policies

In Case of Emergency or Instructional Disruption

Visit these resources for more information on what to do to prepare for emergencies and/or what to do in case of an emergency or instructional disruption:

- Sign up for [Zot Alerts - Emergency Alert System](#)
- [Common Tech Troubles – Tips & Tricks](#)
- [Pedagogical Continuity](#)

Inclusive Excellence

[Student Services and Campus Ally and Awareness Resources](#) are provided by the [Office of Inclusive Excellence](#). UCI has taken a university-wide approach to diversity, equity, and inclusion. This charge is about dismantling the systemic barriers of racism and discrimination that have too long been the norm in higher learning and our broader communities. The Office of Inclusive Excellence serves the campus priorities and is committed to a transformative, whole university effort grounded in our values: expect equity, support diversity, practice inclusion, and honor free speech.

The University of California, Irvine's commitment to ensuring equality and valuing diversity. UCI recognizes that our individual differences can deepen our understanding of one another and the world around us, rather than divide us. In this class, people of all ethnicities, genders and gender identities, religions, ages, sexual orientations, disabilities, socioeconomic backgrounds, regions, and nationalities are strongly encouraged to share their rich array of perspectives and experiences. If you feel your differences may in some way isolate you from UCI's community or if you have a need for any specific accommodations, please speak with the instructor early in the quarter about your concerns and what we can do together to help you become an active and engaged member of our class and community. For some of you, the following resources may be helpful:

- [Diversity Resources](#)
- [Disability Services Center](#)
- [LGBT Resource Center](#)
- [Undocumented Student Support](#)
- [Counseling Center](#)
- [Campus Social Worker](#)
- [Transfer Student Center](#)

Diversity Statement

The University of California, Irvine, in accordance with applicable federal and state law and university policy, does not discriminate on the basis of race, color, national origin, religion, sex, gender identity, pregnancy, physical or mental disability, medical condition, ancestry, marital status, age, sexual orientation, citizenship, or service in the uniformed services. The university

also prohibits sexual harassment. This nondiscrimination policy covers admission, access, and treatment in university programs and activities.

Disability and Accessibility Accommodations *

The University of California, Irvine, is committed to providing a barrier-free environment for learning and an electronic environment that is accessible to everyone, including individuals with disabilities. If you have a disability and feel you need accommodations in this program or a course, please contact the [Disability Services Center](#) (DSC). DSC-approved accommodations will be provided for students who present a Faculty Notification Letter from the DSC.

Class Withdrawal Policy

It is the student's responsibility to officially drop/withdraw from any courses before the deadline posted by the university's registrar's office. Please refer to [UCI's academic calendar](#) for the withdrawal policy, procedure, and refund schedule.

The deadline to add/drop/change is the end of Week 2 by 5:00 PM. The rules/procedures for making changes after the second week are described on [the UCI University Registrar's Withdrawals webpage](#).

Academic Integrity *

The University of California, Irvine, is an institution of learning, research, and scholarship that is strengthened by the existence of an environment of integrity. As members of the academic community, students are responsible for maintaining this environment, subscribe to the practice of academic integrity, and accept individual responsibility for their work and actions. Violations of academic integrity are unacceptable and will not be tolerated because they devalue the teaching and learning experience for the entire community. Observing basic honesty in one's work, words, ideas, and actions is a principle to which all members of the community are required to subscribe.

In most cases, students violating academic integrity will get an "F" for the course, and in extreme cases, they may be expelled from the university. Please inspect the [Academic Integrity Information for Students](#) for more detail. Examples of cheating are copying exams from other students, stealing exam keys from the TA or instructor, or paying someone to take the exam for you. Please do not jeopardize your future career by being irresponsible. If you are unsure whether something you are planning on doing may be considered cheating, please come and speak with me before doing it. "I didn't know that was cheating" will not be considered an acceptable excuse.

Third-Party Tools

Please click here for [information regarding the accessibility statements and the privacy statements](#).

Copyrights *

This course is provided by The University of California, which has [policies regarding copyright](#). Materials used in connection with this course may be subject to copyright protection. Refer to the information provided in each video/file/module/unit for copyright information for each work. The course content related video/file/module/unit was created to be used in compliance with the TEACH Act. 17 U.S.C. §110(2).

My lectures and course materials, including PowerPoint presentations, tests, outlines, and similar materials, are protected by U.S. copyright law and by [University policy](#). I am the exclusive owner of the copyright in those materials that I create. You may take notes and make copies of course materials for your own use. You may also share those materials with other students who are registered and enrolled in this course.

As seen in the [university copyright policy](#), you may not reproduce, distribute or display (digitally post/upload) lecture notes or recordings or course materials in any other way—whether or not a fee is charged—without my express written consent. You also may not allow others to do so.

If you do so, you may be subject to student conduct proceedings under the [Code of Student Conduct, Section 102.23](#).

Similarly, you own the copyright in your original papers and exam essays. If I am interested in posting your answers or papers on the course web site, I will ask for your written permission.

This syllabus is subject to change at any time.